Detailed Methodology: Biologically-Inspired Patch Attention (BIPA) Transformer (End-to-End Training)

Abstract

This document details the methodology for the Biologically-Inspired Patch Attention (BIPA) transformer, specifically adapted for training with only **image-level classification labels**. This approach replaces the reliance on segmentation data by integrating an **Adaptive BAM Generator (A-BAMNet)** that is optimized jointly with the main transformer in an end-to-end fashion. The A-BAMNet learns to discover and highlight species-discriminative biological features implicitly, akin to an Attention-based Multiple Instance Learning strategy.

1 Introduction to BIPA

The Biologically-Inspired Patch Attention (BIPA) transformer is proposed to address limitations in standard Vision Transformers (ViTs) when dealing with fine-grained visual classification tasks. ViTs typically apply uniform attention across all image patches, potentially overlooking subtle yet crucial features. BIPA introduces a novel component: a Biologically-Inspired Attention Map (BAM), which guides the transformer's attention towards regions containing discriminative biological markers (e.g., wing venation, leg banding). This explicit guidance aims to improve feature extraction and classification accuracy.

2 BIPA Architecture Overview

The BIPA architecture can be conceptualized in three main stages:

1. **Micro-Patch Embedding**: Segmentation of the input image into small micro-patches and their linear projection into patch tokens.
2. **Biological Attention Map (BAM) Generation**: Creation of a differentiable, learned attention map highlighting biologically significant regions, trained end-to-end.
3. **Weighted Self-Attention (BIPA Attention)**: Integration of the BAM into the transformer's self-attention mechanism to modulate attention weights.

3 Detailed Methodology

3.1 Pre-processing

Mosquito localization using some Yolo (latest) for the speedy localization of mosquitoes. The image $\mathbf{I}$ is the cropped region containing the mosquito.

3.2 Micro-Patch Embedding

The first step involves preparing the input image for the transformer encoder.

(a) **Input Image Division**: An input image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ (Height, Width, Channels) is divided into a grid of non-overlapping micro-patches $\mathbf{p}_{i,j} \in \mathbb{R}^{P \times P \times C}$, where P is the micro-patch size (e.g., 8×8 pixels). This results in $N = (H/P) \times (W/P)$ micro-patches.

(b) **Flattening**: Each micro-patch $\mathbf{p}_k$ (where $k = 1, \ldots, N$) is flattened into a 1D vector:

$$\mathbf{p}'_k \in \mathbb{R}^{P^2C}$$

(c) **Linear Projection**: These flattened patches are linearly projected into a d_{model}-dimensional embedding space:

$$\mathbf{E}_k = \mathbf{p}'_k \mathbf{W}_p + \mathbf{b}_p$$

where $\mathbf{W}_p \in \mathbb{R}^{(P^2C)\times d_{\text{model}}}$ is the learnable projection matrix and $\mathbf{b}_p \in \mathbb{R}^{d_{\text{model}}}$ is the bias vector.

(d) **Positional Embeddings**: To retain spatial information, learnable positional embeddings $\mathbf{P}_k \in \mathbb{R}^{d_{\text{model}}}$ are added to the patch embeddings:

$$\mathbf{T}_k = \mathbf{E}_k + \mathbf{P}_k$$

The sequence of these patch tokens, along with an optional learnable `<cls>` token, forms the input to the BIPA encoder:

$$\mathbf{T} = [\mathbf{t}_{\text{cls}}, \mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_N] \in \mathbb{R}^{(N+1)\times d_{\text{model}}}$$

3.3 Biological Attention Map (BAM) Generation (End-to-End Learning)

The $\mathbf{M}_B$ is generated by an **Adaptive BAM Generator (A-BAMNet)**, which is trained implicitly using the downstream classification loss.

$$\mathbf{M}_B = \text{A-BAMNet}(\mathbf{I}; \theta_{\text{BAM}})$$

3.3.1 Heuristic BAM Generator (Initialization/Feature-based Augmentation)

The heuristic approach remains valuable for initializing the A-BAMNet or for generating fixed feature maps that can be concatenated with $\mathbf{I}$ to help the A-BAMNet focus on biological cues.

(a) **Feature-Specific Filters**: Apply filters (Gabor, LBP, LoG, etc.) to the input image $\mathbf{I}$ to generate feature maps $\{M_V, M_B, M_A, \ldots\}$.

(b) **Feature Aggregation**: Aggregate the feature maps into M_{raw}.

$$M_{raw}(x, y) = \text{Aggregate}(M_V(x, y), M_B(x, y), M_A(x, y), \ldots)$$

This M_{raw} can be used as an **additional channel input** to the A-BAMNet, or the derived $\mathbf{M}_{B\text{Heuristic}}$ can serve as a strong initialization point for the A-BAMNet weights θ_{BAM}.

3.3.2 Adaptive BAM Generator (A-BAMNet)

The A-BAMNet is a lightweight, differentiable module trained jointly with the BIPA encoder.

(a) **A-BAMNet Architecture**: A lightweight CNN (e.g., using a few 3×3 convolutional layers followed by a feature map downscaling or simple attention head) is used. It must be designed to output a single-channel spatial map.

$$M_{\text{raw}} = \text{A-BAMNet}(\mathbf{I}) \in \mathbb{R}^{H' \times W'}$$

(b) **Patch-level Averaging and Normalization**: The raw map M_{raw} is aligned with the transformer patches and normalized using a non-linear activation. We prefer Sigmoid for flexibility.

$$\mathbf{M}_{Bk} = \frac{1}{|\mathbf{p}_k|} \sum_{(x,y)\in\mathbf{p}_k} M_{\text{raw}}(x, y)$$

$$\mathbf{M}_B = \text{Sigmoid}(\mathbf{M}_B) \in \mathbb{R}^N$$

3.3.3 End-to-End Joint Training and Loss

The A-BAMNet weights (θ_{BAM}) and the BIPA encoder weights (θ_{TRF}) are optimized together using only the image-level classification loss. This is an instance of **Attention-based Multiple Instance Learning (MIL)** where the $\mathbf{M}_B$ acts as the implicit instance attention weight.

(a) **Prediction**: The final prediction $\hat{y}$ is derived from the `<cls>` token after L BIPA encoder blocks.

$$\hat{y} = \text{Classifier}(\text{BIPAEncoder}(\mathbf{T}, \mathbf{M}_B(\mathbf{I}; \theta_{\text{BAM}}); \theta_{\text{TRF}}))$$

(b) **Loss Function**: The total loss is the Cross-Entropy loss $\mathcal{L}_{\text{CLS}}$ between the prediction $\hat{y}$ and the ground truth label y.

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{CLS}}(y, \hat{y})$$

(c) **Optimization**: The gradients are backpropagated through the BIPA attention mechanism and, crucially, through the A-BAMNet.

$$\min_{\theta_{\text{BAM}}, \theta_{\text{TRF}}} \mathcal{L}_{\text{Total}}$$

This forces the A-BAMNet to generate high $\mathbf{M}_B$ scores for patches (instances) containing features that minimize the classification error $\mathcal{L}_{\text{CLS}}$, implicitly discovering the biological features without explicit segmentation labels.

3.4 Weighted Self-Attention (BIPA Attention Layer)

The generated $\mathbf{M}_B$ is integrated into the multi-head self-attention (MHSA) mechanism of the transformer encoder. Each BIPA attention block consists of:

(a) **Query, Key, Value Projections**: For each head h in a MHSA block, the input patch tokens $\mathbf{T}$ are linearly projected into Query ($\mathbf{Q}_h$), Key ($\mathbf{K}_h$), and Value ($\mathbf{V}_h$) matrices:

$$\mathbf{Q}_h = \mathbf{T}\mathbf{W}_Q^{(h)}$$

$$\mathbf{K}_h = \mathbf{T}\mathbf{W}_K^{(h)}$$

$$\mathbf{V}_h = \mathbf{T}\mathbf{W}_V^{(h)}$$

where $\mathbf{W}_Q^{(h)}, \mathbf{W}_K^{(h)}, \mathbf{W}_V^{(h)} \in \mathbb{R}^{d_{\text{model}} \times d_k}$ are learnable weight matrices for head h, and $d_k = d_{\text{model}}/\text{num_heads}$.

(b) **Attention Score Calculation with BAM Modulation**: The dot-product attention scores are computed. Let $\hat{\mathbf{M}}_B \in \mathbb{R}^{(N+1)\times(N+1)}$ be the expanded $\mathbf{M}_B$ where $\hat{\mathbf{M}}_{Bi,j} = \mathbf{M}_{Bj}$.

The attention scores are modified before the softmax using the **Additive Modulation (Recommended)**:

$$\text{Scores}_h = \frac{\mathbf{Q}_h \mathbf{K}_h^T}{\sqrt{d_k}}$$

$$\text{Scores}_{h,\text{BIPA}} = \text{Scores}_h + \alpha \cdot \hat{\mathbf{M}}_B$$

Here, α is a **learnable scalar weight** that controls the global influence of the learned BAM bias.

(c) **Softmax and Weighted Sum**:

$$\mathbf{A}_h = \text{softmax}(\text{Scores}_{h,\text{BIPA}})$$

$$\text{WeightedValue}_h = \mathbf{A}_h \mathbf{V}_h$$

(d) **Concatenation and Final Projection**:

$$\text{MultiHeadOutput} = \text{Concat}(\text{WeightedValue}_1, \ldots, \text{WeightedValue}_{\text{num_heads}})\mathbf{W}_O$$

where $\mathbf{W}_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$.

3.5 BIPA Encoder Block Structure

Each BIPA encoder block follows the standard transformer architecture but replaces the conventional MHSA with the BIPA Attention mechanism.

(a) **Layer Normalization (Pre-Norm)**:

$$\mathbf{Z}_1 = \text{BIPAAttention}(\text{LayerNorm}(\mathbf{T})) + \mathbf{T}$$

(b) **Feed-Forward Network (FFN)**:

$$\mathbf{Z}_2 = \text{MLP}(\text{LayerNorm}(\mathbf{Z}_1)) + \mathbf{Z}_1$$

This sequence is repeated for L encoder blocks. The final output of the transformer encoder (typically the `<cls>` token representation) is then passed to a classification head for species prediction.

4 Novelty and Similarity to Available Transformers (Updated)

4.1 Novelty of BIPA

The primary novelty of BIPA, particularly in the end-to-end configuration, lies in its **implicit learning of biological attention using MIL principles**.

- **Implicit Biological Guidance (Key Novelty)**: The A-BAMNet learns to generate the biological saliency map *solely* from the classification signal. This avoids the cost of segmentation labeling while ensuring the guidance remains biologically relevant, as the model is forced to focus on the features necessary for species discrimination.
- **Differentiable Attention Bias**: The $\mathbf{M}_B$ is a differentiable, content-aware bias added directly to the attention logits ($\frac{QK^T}{\sqrt{d_k}}$). This is a strong, structured injection of prior knowledge/learned saliency, making it more effective than simply masking patches or post-hoc analysis.
- **Micro-Patch Resolution**: The use of micro-patches (e.g., 8×8) ensures that the subtle biological features (venation, banding) remain resolvable, and the learned $\mathbf{M}_B$ can assign fine-grained importance scores to these specific regions.

4.2 Similarity to Available Transformers

BIPA shares foundational elements with:

- **Vision Transformer (ViT)**: Provides the core patching, embedding, and encoder structure.
- **Attention-based Multiple Instance Learning (MIL)**: The mechanism for training the A-BAMNet relies heavily on the principles of MIL, where the image-level label is used to discover the most influential 'instances' (patches). BIPA generalizes this by integrating the MIL-derived attention map directly as a **bias** within the transformer's own attention score calculation.
- **Saliency-Guided/Attention Transfer Models**: Similar to models that use external attention, but BIPA makes the attention generator *internal* and *jointly optimized* with the transformer, coupling the feature discovery and feature processing into one cohesive, end-to-end framework.

References

[1] Dosovitskiy, A., Beyer, L., Koll F., et al. (2020). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *arXiv preprint arXiv:2010.11929*.